LRR-Net: An Interpretable Deep Unfolding Network for Hyperspectral Anomaly Detection

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***Abstract*— Considerable endeavors have been expended toward enhancing the representation performance for hyperspectral anomaly detection (HAD) through physical model-based methods and recent deep learning-based approaches. Of these methods, the low-rank representation (LRR) model is widely adopted for its formidable separation capabilities for background and target features, however, its practical applications are limited due to the reliance on manual parameter selection and subpar generalization performance. To this end, this article presents a new HAD baseline network, referred to as LRR-Net, which synergizes the LRR model with deep learning techniques. LRR-Net leverages the alternating direction method of multipliers (ADMM) optimizer to solve the LRR model efficiently and incorporates the solution as prior knowledge into the deep network to guide the optimization of parameters. Moreover, LRR-Net transforms the regularized parameters into trainable parameters of the deep neural network, thus alleviating the need for manual parameter tuning. Additionally, this article proposes a sparse neural network embedding to demonstrate the scalability of the LRR-Net framework. Empirical evaluations on eight distinct datasets illustrate the efficacy and superiority of the proposed approach compared to the state-of-the-art methods.**

***Index Terms*— Alternating direction method of multipliers (ADMM), anomaly detection, artificial intelligence, deep unfolding, hyperspectral image, interpretability, low-rank representation (LRR), sparse representation.**

I. Introduction

HYPERSPECTRAL imagery (HSI) has been proven to be a powerful tool in military and civilian applications that can be used to detect different objects based on their spectral features [1], [2], [3], [4]. HSI provides rich spectral information with hundreds or even thousands of bands [5], enabling high-precision detection of targets of interest. However, due to complex imaging environments, low spatial resolution, and the absence of prior information about anomalies [6], the hyperspectral anomaly detection (HAD) task is often treated as an unsupervised problem, aimed at separating background and anomalous targets from observed data based on specific features. HAD methods can be divided into two main categories: model-based methods and deep learning-based methods, depending on the data-driven approach adopted.

In the early stages of HAD, model-based methods that relied on statistical calculations were widely used. The Reed–Xiaoli (RX) algorithm [7], proposed in 1990, was a typical representative of these methods. It assumed that the background pixels followed a Gaussian multivariate distribution and estimated the background covariance matrix using the square value of the Mahalanobis distance between the target pixels and the background pixels to identify anomalies. This led to the development of several improved methods based on the RX detector, including the kernel-RX (KRX) [8], [9], regularized RX (R-RXD) [10], segmented-RX [11], local-RX (LRX) [12], and weighted-RX [13]. The KRX model represented the background distribution in high-dimensional feature space using a kernel function, while the R-RXD regularized the covariance matrix estimated from all HSI pixels. The segmented-RX used a clustering of image pixels to estimate the background, while the LRX used a double-window structure to estimate the background based on local statistics. The weighted-RX aimed at improving the background information estimation by weighting background pixels more than target pixels. However, in real-world applications, the statistical distribution assumptions made by these methods are often not ideal due to complex landcover distributions [14].

To overcome the limitations of distributional assumptions in HAD, recent years have seen increased interest in representation-based methods. One such method is low-rank representation (LRR) [15], which aims to find the lowest rank representation of all pixels jointly. The anomalies can then be obtained by subtracting the recovered background from the original image using the lowest rank representation. This method characterizes the relationships among all HSI pixels from a global perspective. In addition to the global perspective, the local structure of each pixel's coefficient is crucial for

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accurate representation. To capture this structure, a sparsity criterion is designed in the proposed LRASR method [16]. The background dictionary also plays a significant role in representation power. Another example is the collaborative-representation-based detector [17], which is based on the idea that each background pixel can be approximately represented by its spatial neighbors, while anomalies cannot. However, the development of model-based HAD technology is somewhat limited as follows:

1) the search process for the optimal model solution often requires hundreds of iterations or more;
2) the determination of model parameters requires manual intervention for adjustment; and
3) the algorithm lacks generalization, and once the input image changes, the parameters have to be readjusted.

In recent years, the field of image reconstruction has seen a surge of interest in neural networks, leading to the development of various deep-learning-based methods such as convolutional neural networks (CNNs) [18], [19], autoencoders (AEs) [20], generative adversarial networks (GANs) [21], recurrent neural network (RNN) [22], and long short-term memory (LSTM) [23]. CNN can learn complex layered representations from hyperspectral images, where the latter convolutional layer can be used to extract data features from the previous convolutional layer by convolutional kernel operations, thus enabling the extraction of high-level feature information from low-level feature information [24], [25], [26]. Among unsupervised HAD methods, AEs and their variants [27], [28], [29] are the core of existing methods and are usually more popular, which have been applied to HAD, mainly as preprocessing for detection, i.e., feature extraction or using the residuals between the reconstructed background image and the original image to detect anomalies. The next category is GAN, which generally speaking combines AE and GAN to form an adversarial generative network (AAE), adding a discriminator to the AE [28], [30], [31], [32]. RNN can be used to model the temporal dependencies in hyperspectral data and detect anomalies in the temporal domain [33]. The LSTM is a recursive neural network that captures the time-series information of the data, which can be used for anomaly detection in the time series of the hyperspectral data to reduce abnormal pollution [34].

Compared with the iterative optimization of the physical model-based algorithms, these classical deep neural networks, typically trained using back-propagation algorithms, can more accurately capture and represent the deeper abstraction and complexity of an image, which have proven effective in solving HAD challenges. Nevertheless, the black-box characteristic of these network models limits their interpretability and theoretical guarantees, which can result in undesired effects. Specifically, first, there is a lack of principled approaches for designing network architectures, particularly for the encoder network. Second, neural networks can be difficult to interpret, making it challenging to incorporate prior knowledge into the design of network architecture. Third, deep learning approaches generally require large training dataset sizes, a considerable number of learning parameters, and lengthy training periods to achieve satisfactory performance. To avoid these entirely data-driven end-to-end manners that may have risks for some undesired effects, it may be beneficial to integrate the prior knowledge and the structure of the operators.

In this article, a novel HAD baseline network LRR-Net is proposed, in which the iterative process of solving the LRR model is deeply unfolded. Specifically, the effective search direction of the optimal solution is guided by coupling the traditional model into the deep neural network as prior knowledge, which reduces the dependence on samples, considering that the LRR model has a powerful separation capability and can well detect anomalous targets in complex backgrounds. On the other hand, an explicit process of HAD in an end-to-end fashion is provided by equivalently converting the regularized parameters into a set of learnable deep neural network parameters through precise step-wise correspondence between traditional modules and algorithm operators. In addition, given that alternating direction multiplication (ADMM) [35], [36], [37] is a very powerful solver for the optimization problem with sufficient accuracy and fast convergence rate, we present two novel network architectures for HAD, baseline network, and embedded sparse extended network, respectively. The main contributions of this article can be therefore summarized as follows.

1) We propose an interpretable deep network by fully unfolding the LRR model for the first time, named LRR-Net, for the HAD task. LRR-Net can not only provide the explicit process of HAD in an end-to-end trainable network fashion but also adaptively learn the regularization parameters instead of the previous handcraft design.
2) We extend the LRR-Net by considering the joint low-rank and sparse priors, yielding the LRR-Net-S. LRR-Net-S further enhances the learning ability for anomalous pixels in hyperspectral images by embedding the sparsity.
3) Beyond LADMAP [38], we provide an effective solution with the well-designed ADMM for the proposed LRR-Net and LRR-Net-S. Extensive experiments conducted on a series of widely used hyperspectral datasets demonstrate the superiority of the proposed methods both visually and quantitatively.

The rest of this article is organized as follows. We first introduce in Section II how the ADMM algorithm can be used to obtain the background related to the closed-form solution of the LRR model. Then, Section III describes the first version of our unfolding network for HAD and focuses on the development of the LRR-Net-sparse version and its optimal solution procedure. Extensive experimental results are reported and discussed in Section IV. Finally, we give conclusions and future perspectives in Section V.

II. Brief Review of LRR and Deep Unfolding

In this section, we will review related works on the widely studied LRR model and its optimal solution method using the ADMM method. Additionally, we will introduce the unfolding methods for designing deep architectures.

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A. LRR Model for HAD

Assume $\mathbf{X} = [x_1, \ldots, x_i, \ldots, x_M] \in \mathbb{R}^{M*B}$ is a spectrum measured HSI with $M = C*H$ pixels, where x_i, $(i = 1, \ldots, M)$ is the ith spectral vector with B dimensions. Consider that those background pixels usually have strong between them, while the anomalous pixels are different. This means that the matrix $\mathbf{X}$ can be decomposed into a background part and an anomalous part as follows:

$$\mathbf{X} = \mathbf{DL} + \mathbf{S} \tag{1}$$

where $\mathbf{D}$ is the background dictionary, which is composed of the background samples, and $\mathbf{L}$ is the representation coefficients. $\mathbf{DL}$ denotes the background part and $\mathbf{S}$ denotes the anomalous part. The anomalous part $\mathbf{S}$ is then optimized by solving the following energy minimization problem:

$$\min_{\mathbf{L},\mathbf{S}} ||\mathbf{L}||_* + \beta||\mathbf{S}||_{2,1}, \quad \text{s.t. } \mathbf{X} = \mathbf{DL} + \mathbf{S} \tag{2}$$

where $||\mathbf{L}||_*$ represents the nuclear norm and $||\mathbf{S}||_{2,1} = \sum_{i=1}^{B}(\sum_{j=1}^{M}(s_{i,j})^2)^{1/2}$ denotes $l_{2,1}$-norm of the anomaly matrix $\mathbf{S}$. Furthermore, $\beta \geq 0$ is a regularization parameter that controls the solutions of grouping and low-rank properties.

By introducing an auxiliary variable $\mathbf{J}$ to replace $\mathbf{L}$ such that $\mathbf{L} = \mathbf{J}$ to make the objective function separable. The augmented Lagrangian function corresponding to the objective function can be obtained

$$\begin{aligned}\mathscr{L}_U(\mathbf{J}, \mathbf{S}, \mathbf{L}, \mathbf{P}_1, \mathbf{P}_2) = {} & \|\mathbf{J}\|_* + \beta\|\mathbf{S}\|_{2,1} \\ & + \frac{\mu}{2}\|\mathbf{J} - \mathbf{L} - \mathbf{P}_2/\mu\|_{\mathrm{F}}^2 \\ & + \frac{1}{2}\|\mathbf{S} - (\mathbf{X} - \mathbf{DL} - \mathbf{P}_1/\mu\|_{\mathrm{F}}^2 \\ & - \frac{1}{2\mu}\left(\|\mathbf{P}_1\|_{\mathrm{F}}^2 + \|\mathbf{P}_2\|_{\mathrm{F}}^2\right)\end{aligned} \tag{3}$$

where $\mathbf{P}_1 > 0$ and $\mathbf{P}_2 > 0$ are two penalty parameters, and $\|\cdot\|_F$ refers to the Frobenius norm.

Problem (3) can be solved with the alternating direction method by alternatively updating only one variable at a time while fixing the others. During the kth iteration, the problem can be divided into the following steps.

1) ***J** Step:* Fix $\mathbf{S}$, $\mathbf{L}$, $\mathbf{P}_1$, and $\mathbf{P}_2$ to update $\mathbf{J}$; the objective function can be written as follows:

$$\mathbf{J}^{(k)} = \underset{\mathbf{J}}{\operatorname{argmin}} ||\mathbf{J}||_* + \frac{\mu}{2}\left\|\mathbf{J}^{(k-1)} - \mathbf{L}^{(k-1)} - \mathbf{P}_2^{(k-1)}/\mu\right\|_{\mathrm{F}}^2. \tag{4}$$

2) ***S** Step:* Fix $\mathbf{J}$ and $\mathbf{L}$, $\mathbf{P}_1$, and $\mathbf{P}_2$ to update $\mathbf{S}$; the objective function can be written as follows:

$$\begin{aligned}\mathbf{S}^{(k)} = \underset{\mathbf{S}}{\operatorname{argmin}}\ & \beta||\mathbf{S}||_{2,1} + \left\|\mathbf{S}^{(k-1)}\right\| \\ & - \left(\mathbf{X} - \mathbf{DL}^{(k-1)} - \mathbf{P}_1^{(k-1)}\Big/\mu\right)_{\mathrm{F}}^2.\end{aligned} \tag{5}$$

3) ***L** Step:* Fix $\mathbf{J}$, $\mathbf{S}$, $\mathbf{P}_1$, and $\mathbf{P}_2$ to update $\mathbf{L}$; the objective function can be written as follows:

$$\begin{aligned}\mathbf{L}^{(k)} = \underset{\mathbf{L}}{\operatorname{argmin}}\ & \frac{\mu}{2}\left\|\mathbf{J}^{(k-1)} - \mathbf{L}^{(k-1)} - \mathbf{P}_2^{(k-1)}/\mu\right\|_{\mathrm{F}}^2 \\ & + \left\|\mathbf{S}^{(k-1)} - (\mathbf{X} - \mathbf{DL}^{(k-1)} - \mathbf{P}_1^{(k-1)}/\mu)\right\|_{\mathrm{F}}^2.\end{aligned} \tag{6}$$

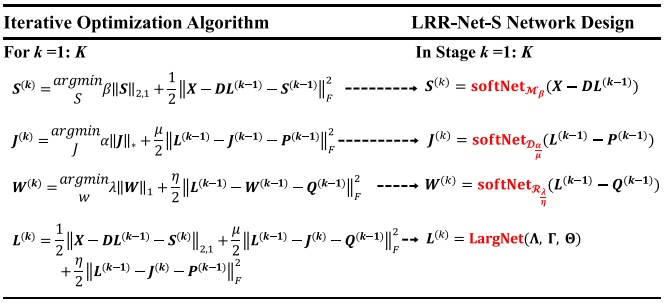

Fig. 1. Illustration of the relationship between the iterative optimization algorithm and the network structure.

4) ***P** Step:* Fix $\mathbf{J}$, $\mathbf{S}$, and $\mathbf{L}$ to update $\mathbf{P}_1$ and $\mathbf{P}_2$; the objective function can be written as follows:

$$\begin{aligned}\mathbf{P}_1^{(k)} &= \mathbf{P}_1^{(k-1)} - \mu(\mathbf{X} - \mathbf{DL}^{(k)} - \mathbf{S}^{(k)}) \\ \mathbf{P}_2^{(k)} &= \mathbf{P}_2^{(k-1)} - \mu(\mathbf{L}^{(k)} - \mathbf{J}^{(k)}).\end{aligned} \tag{7}$$

Once the optimization is finished, the anomaly matrix $\mathbf{S}$ can be extracted, and the anomaly value for the ith spectrum $\mathbf{X}_i$ can be determined as $||\mathbf{S}||_{2,1}$, where $\mathbf{S}_i$ is the ith column of anomaly matrix $\mathbf{S}$.

B. Deep Unfolding

Albeit the initial improvement of the intelligence degree has been achieved for HAD through deep learning technology, they still use the architectures, including CNN or AE, as the main module to build the network, which thus still lacks sufficient interpretability.

In recent years, the unfolding or unrolling technique has emerged as a potential solution to design the interpretable network, which has been applied to unmixing [39], [40], denoising [41] and fusion [42] in hyperspectral images. In unfolding networks, the basic structure is derived from the iterative updating process of the model-based algorithm, and the learnable parameters in the network are automatically obtained through training. Specifically, with these techniques, the target task is first mapped onto an optimization problem, which is usually solved by an iterative solver. Then, each step in the iterative solver could be converted to a network operation. Finally, the network is constructed by concatenating several iterations or layers in the network of such an operation [43].

From the above, it can be seen that the unfolding network can be seen as a combination of model-driven and data-driven techniques, which can inherit the advantages of both techniques. Yang et al. [44] show that the network unfolded from an iterative algorithm can achieve better performance than the original iterative algorithm. This naturally motivates us to investigate a general but interesting question—can the performance of AD be further improved by incorporating the advantages of the model-driven prior and the data-driven, which can automatically learn the discriminative features?

III. LRR-Net: LRR Meets Deep Unfolding

In this section, considering the generalized deep unfolding network for HAD, we use a different ADMM iterative solution

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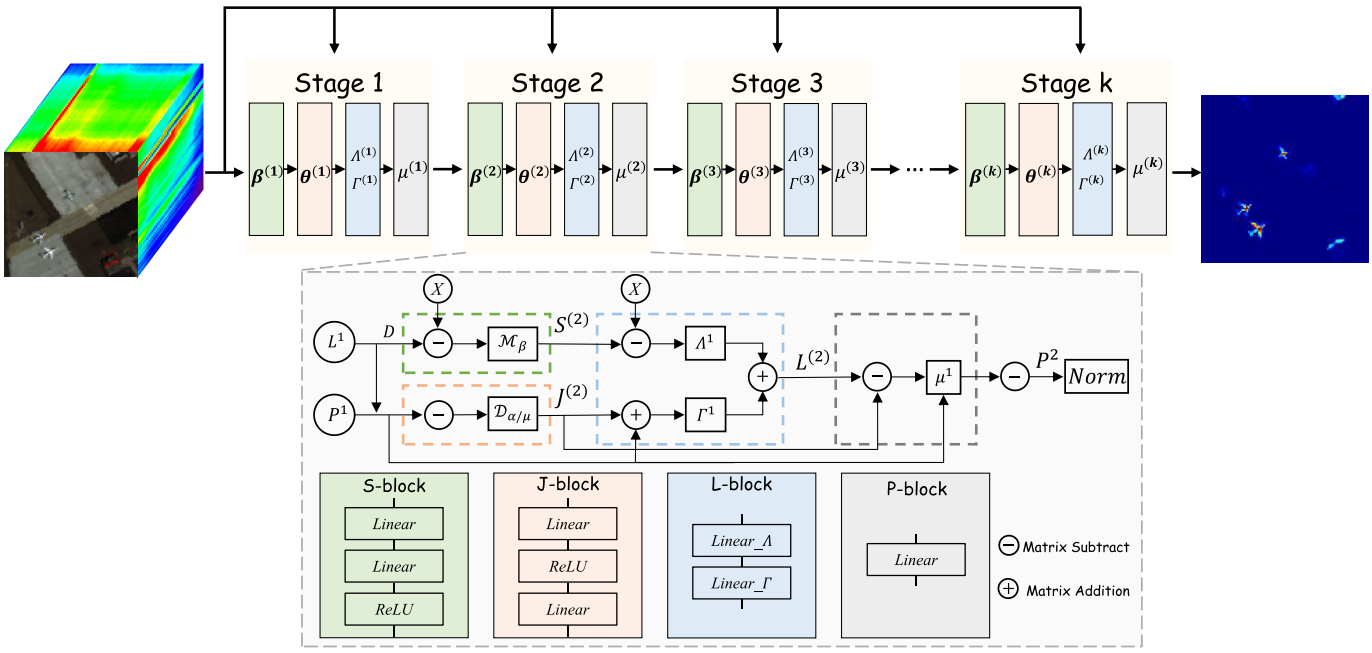

Fig. 2. Overview of the proposed network architecture (LRR-Net) for HAD.

method to solve the LRR model and unfold the process to design a base version of LRR-Net. Then, we propose the LRR-Net-sparse version by embedding the sparse module in LRR-Net and derive its solution procedure illustrating the scalability of the base version.

A. Method Overview

Commonly, a satisfactory result needs to run the ADMM algorithm for hundreds of iterations, and the process requires manual tuning. On the other hand, the black-box operation of deep learning makes the processing of the HAD task unstable. To overcome these difficulties, we tried to define a baseline network LRR-Net by deep unfolding for HAD task to learn all the above functions and parameters from the data-based.

Our LRR-Net is defined by unfolding and generalizing the iterative ADMM algorithm to be a deep architecture. We first map ADMM iterative procedures in (8) and (2) to a data flow graph respectively shown in Fig. 2. This graph comprises nodes corresponding to different operations in ADMM iterations, and directed edges corresponding to the data flow between nodes (i.e., operations). In this case, the nth iteration of the ADMM algorithm corresponds to the nth stage of the data flow graph. Thus, given measurement data, LRR-Net converts these parameters into a series of learnable parameters, and the to-be-optimized variables can be updated through strict closed-form solutions, which reduces the extreme dependence of the model on manual parameter selection and further improves the generalization ability.

B. LRR-Net: Deep Unfolding Solver Into Neural Network

In the study of anomaly detection, the mentioned model optimization strategy [16] is widely used, and we used the same construction method for dictionary $\mathbf{D}$ obtained by K-Means clustering. The difference is that we used fewer constraints to describe the optimization model for (2) inspired by the unmixed model to make the model more generalizable. Thus, good surrogate optimization is provided as follows:

$$\min_{\mathbf{L},\mathbf{S}} \frac{1}{2}||\mathbf{X}-\mathbf{D}\mathbf{L}-\mathbf{S}||_{\mathrm{F}}^2+\alpha||\mathbf{L}||_*+\beta||\mathbf{S}||_{2,1} \tag{8}$$

where $\alpha \geq 0$ and $\beta \geq 0$ are two regularization parameters that control the solution of the low-rank properties.

The augmented Lagrangian function of (8) can be written as

$$\mathscr{L}_U(\mathbf{J},\mathbf{S},\mathbf{L},\mathbf{P}_j)=\frac{1}{2}\|\mathbf{X}-\mathbf{D}\mathbf{L}-\mathbf{S}\|_{\mathrm{F}}^2+\frac{\mu}{2}\|\mathbf{L}-\mathbf{J}+\mathbf{P}\|_{\mathrm{F}}^2 +\beta\|\mathbf{S}\|_{2,1}+\alpha\|\mathbf{J}\|_*+l_R^+(\mathbf{J}) \tag{9}$$

where $\mathbf{J}$ is an auxiliary variable, which is used as a split variable to replace $\mathbf{L}$, and $\mathbf{P}>0$ is a penalty parameter. $(\cdot)^+$ denotes an operator that intercepts the positive part of each component of the matrix, and $l_R^+(\mathbf{J})$ is defined as $\mathbf{J}\succeq\mathbf{0}$.

Equation (9) can be divided into the following sub-problem:

$$\begin{cases} \underset{\mathbf{S}}{\operatorname{argmin}}\ \beta||\mathbf{S}||_{2,1}+\frac{1}{2}||\mathbf{X}-\mathbf{D}\mathbf{L}-\mathbf{S}||_{\mathrm{F}}^2 & (10)\\ \underset{\mathbf{J}}{\operatorname{argmin}}\ \alpha||\mathbf{J}||_*+\frac{\mu}{2}||\mathbf{L}-\mathbf{J}-\mathbf{P}||_{\mathrm{F}}^2 & (11)\\ \underset{\mathbf{L}}{\operatorname{argmin}}\ \frac{1}{2}||\mathbf{X}-\mathbf{D}\mathbf{L}-\mathbf{S}||_{\mathrm{F}}^2+\frac{\mu}{2}||\mathbf{L}-\mathbf{J}-\mathbf{P}||_{\mathrm{F}}^2 & (12)\\ \mathbf{P}\leftarrow\mathbf{P}-\mu(\mathbf{L}-\mathbf{J}) & (13)\end{cases}$$

where μ is an update rate for the Lagrangian Multiplier.

We mentioned earlier that each iteration block in the deeply expanded network can be considered as a neural network layer corresponding to one iteration of the ADMM algorithm in Fig. 1. The model parameters $\{\Lambda,\Gamma,\alpha,\mu,\beta\}$ associated with the ADMM algorithm, which are manually tuned for the

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Algorithm 1 LRR-Net

Input: Observed HSI $\mathbf{X}$, $\mathbf{D}$, $\mathbf{L}_0$, $\mathbf{S}_0$, and parameters β, μ, η.

Output: Background $\mathbf{L}$, Anomaly map $\mathbf{S}$.

Initialization: $\mathbf{J}=\mathbf{0}$, $\mathbf{P}=\mathbf{0}$, $\Lambda=\mathbf{0}$, $\Gamma=\mathbf{0}$, $\alpha=0.5$, $\beta=10^{-5}$, $\mu=1$.

while $0\le k\le K$ *or meet the convergence conditions* **do**

  Fix other variables to update $\mathbf{S}$ by Eq. (14)

$$\mathbf{S}=\mathbf{softNet}_{\mathcal{M}_\beta}(\mathbf{X}-\mathbf{DL}),$$

  Fix other variables to update $\mathbf{J}$ by Eq. (17)

$$\mathbf{J}=\mathbf{softNet}_{\mathcal{D}_{\alpha/\mu}}(\mathbf{L}-\mathbf{P}).$$

  Fix other variables to update $\mathbf{L}$ by Eq. (21)

$$\mathbf{L}=\mathbf{LargNet}(\Lambda,\Gamma).$$

  Fix other variables to update $\mathbf{P}$ by Eq. (22)

$$\mathbf{P}=\mathbf{Linear}(P_\mu).$$

  Check the convergence conditions:

  if $\|\mathbf{L}-\mathbf{J}\|_{\mathrm{F}}<\varepsilon$, $||x^k-\hat{x}^{k-1}||_F^2<\zeta$ **then**

    Stop iteration;

  else

    $k\leftarrow k+1$;

  end

end

conventional approach, can be replaced by learnable parameters in each neural network layer. Thus, we can construct several different feed-forward neural networks by connecting K iteration blocks to simulate K iterations of the ADMM algorithm.

The framework of the proposed LRR-Net is derived from the previous low-rank model using the deep unfolding approach. Specifically, each layer of the proposed LRR-Net consists of four modules, including the **S**-block, **J**-block, **L**-block, and the **P**-block. The architecture of the proposed LRR-Net is shown in Fig. 2(a), and the details of the proposed LRR-Net are described next.

1) S-Block: Recalling the update process of the low-rank model in the ADMM algorithm in (10), it first needs to calculate the intermediate variable $\mathbf{S}^{(k)}$, which is given by

$$\mathbf{S}^{(k)}=\mathbf{softNet}_{\mathcal{M}_\beta}(\mathbf{X}-\mathbf{DL}^{(k-1)}) \tag{14}$$

where $\mathbf{softNet}_{\mathcal{M}_\beta}(\mathbf{s})$ represents the soft-thresholding operator, which is given as

$$\mathcal{M}_\beta(\mathbf{s})=\mathbf{sign}(\|\mathrm{s}\|_2)\mathbf{max}(0,\|\mathrm{s}\|_2-\beta) \tag{15}$$

where $\|\cdot\|_2$ represents the l_2-norm computation for each column.

From the expression of the soft-thresholding operator, it achieves the same function as the ReLU function. Therefore, the update of the $\mathbf{S}$ in the LRR-Net can be achieved by

$$\frac{\mathbf{X}-\mathbf{DL}}{\|\mathbf{X}-\mathbf{DL}\|_2}\mathrm{ReLU}(\|\mathbf{X}-\mathbf{DL}\|_2-\beta). \tag{16}$$

2) J-Block: Similarly, to update the (11) in the ADMM, it first needs to calculate the intermediate variable $\mathbf{J}^k$ in the current step, which is given as

$$\mathbf{J}^{(k)}=\mathbf{softNet}_{\mathcal{D}_{\alpha/\mu}}(\mathbf{L}^{(k-1)}-\mathbf{P}^{(k-1)}) \tag{17}$$

where $\mathbf{softNet}_{\mathcal{D}_{\alpha/\mu}}(J)$ is solved via the singular value thresholding (SVT) operator [15] as follows.

1) Input a matrix $\mathbf{M}$ of rank r and consider the singular value decomposition (SVD)

$$\mathbf{M}=\mathbf{U}\sum\mathbf{V},\quad \sum=\mathrm{diag}(\{\sigma\}_+). \tag{18}$$

2) For each $\alpha/\mu\ge 0$, we define the operator $\mathcal{D}_{\alpha/\mu}$ as follows:

$$\begin{aligned}\mathcal{D}(\mathbf{M})&:=\mathbf{U}\times\mathcal{D}_{\alpha/\mu}(J)\times\mathbf{V}\\ \mathcal{D}_{\alpha/\mu}(J)&=\mathrm{diag}(\{\sigma_k-\alpha/\mu\}_+)\end{aligned} \tag{19}$$

where $\{\cdot\}_+=\max(0,x)$. Using (19), $\|\mathbf{J}\|_*$ can be computed by $\mathbf{softNet}_{\mathcal{D}_{\alpha/\mu}}(J)$.

Note that the rectified linear unit (ReLU) is a component-wise rectified linear unit operation, which is the same as the proximal operator of the nuclear norm from the process [45]. Therefore, the (22) can be written as

$$\mathbf{U}\times\mathrm{ReLU}(\mathrm{diag}\{(\mathbf{L}-\mathbf{P})-(\alpha/\mu)\mathbf{I}\})\times\mathbf{V}. \tag{20}$$

3) L-Block: This module is derived by unfolding $\mathbf{L}$-*update*, which can write (k)th iterate $\mathbf{L}^{(k)}$ given the $(k-1)$th estimates $\mathbf{S}(k-1)$, $\mathbf{J}(k-1)$, and $\mathbf{P}(k-1)$ along with $\mathbf{X}$ as follows:

$$\mathbf{L}=\mathbf{LargNet}(\Lambda,\Gamma) \tag{21}$$

where $\Lambda=(\mathbf{D}^T\mathbf{D}+\mu\mathbf{I})^{-1}\mathbf{D}^T$, $\Gamma=(\mathbf{D}^T\mathbf{D}+\mu\mathbf{I})^{-1}\mu$, and let $\mathbf{I}$ be a unit matrix. Note that the operation can also be viewed as two distinct linear layers in our network.

4) P-Block: To improve the robustness, we set the step size μ as a learnable parameter to have the following operation:

$$\mathbf{P}^{(k)}=\mathbf{Linear}(P_\mu) \tag{22}$$

where $\mathbf{Linear}(\cdot)$ represents the linear layer corresponding to (13), and P_μ can be calculated by $\mathbf{P}-\mu(\mathbf{L}-\mathbf{J})$, where $\mathbf{J}$ is also a learnable parameter offering additional flexibility, and the parameter μ is updated by $\min(\max\mu,\mu*\beta)$.

Actually, we empirically and experimentally found that the normalization operation performed between the neighboring stages is very useful in the LRR-Net's optimization and convergence. The benefits are twofold. One can enable stable model training, thereby avoiding the gradient disappearance and explosion in the unfolding network training process. It, on the other hand, accelerates the convergence of the learning algorithm, tending to find a better solution.

In both cases, the network takes as input the HSI spectrum X, accompanied by the pseudo inputs L^0 and P^0, which we both set to 0 in line with the standard approach in ADMM algorithms. Then these inputs will go through K network layers with learnable parameters $\{\Lambda,\Gamma,\alpha,\beta,\mu\}_{k=1}^K$.

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C. Sparse Prior in LRR-Net

As seen in [15], the low-rank criterion is superior at capturing the global structure of observed data. However, each spectrum has its own local structure. The more accurate the description of the local structure, the more accurate the representation of the observed data. Therefore, the sparse nature of the matrix allows us to describe the local structure. In this section, we will jointly embed the sparse prior in (8), which is can be written as follows:

$$\min_{\mathbf{L},\mathbf{S}} \ \tfrac{1}{2}||\mathbf{X}-\mathbf{DL}-\mathbf{S}||_{\mathrm{F}}^2+\alpha\|\mathbf{L}\|_*+\lambda\|\mathbf{L}\|_1+\beta\|\mathbf{S}\|_{2,1} \quad (23)$$

compared with (8), (23) imposes a sparse constraint on the background by $||\cdot||_1$, which is the l_1 norm of a matrix. In addition, $\alpha>0$, $\lambda>0$, and $\beta>0$ are the parameters to weigh these constraint limits.

Similarly, by introducing multiple auxiliary variables $\mathbf{J}$ and $\mathbf{W}$ to replace $\mathbf{L}$ such that $\mathbf{L}=\mathbf{J}$, and $\mathbf{W}=\mathbf{J}$, respectively, with two penalty parameters $\mathbf{P}$ and $\mathbf{Q}$. In particular, $(\cdot)^+$ denotes an operator that intercepts the positive part of each component of the matrix, and $l_R^+(\mathbf{J})$ and $l_R^+(\mathbf{W})$ are defined as $\mathbf{J}\succeq\mathbf{0}$ and $\mathbf{W}\succeq\mathbf{0}$, respectively. Equation (23) can be equivalently solved by minimizing the following augmented Lagrangian function:

$$\begin{aligned}\mathscr{L}_U(\mathbf{J},\mathbf{W},\mathbf{S},\mathbf{L},\mathbf{P},\mathbf{Q},\mu,\eta)&=\frac{1}{2}\|\mathbf{X}-\mathbf{DL}-\mathbf{S}\|_{\mathrm{F}}^2\\&+\beta\|\mathbf{S}\|_{2,1}+\alpha\|\mathbf{J}\|_*+\frac{\mu}{2}\|\mathbf{L}-\mathbf{J}+\mathbf{P}\|_{\mathrm{F}}^2+l_R^+(\mathbf{J})\\&+\lambda\|\mathbf{W}\|_1+\frac{\eta}{2}\|\mathbf{L}-\mathbf{J}+\mathbf{Q}\|_{\mathrm{F}}^2+l_R^+(\mathbf{W}).\end{aligned} \quad (24)$$

The specific optimization flow for solving the problem is summarized in Algorithm 2.

Compared to the optimization iteration steps of the LRR-Net version, the three update networks for $\mathbf{S}$, Lagrange multipliers $\mathbf{J}$ and its penalty parameter $\mathbf{P}$ are the same in the LRR-Net-S version, which jointly performs the sparse representation. Next, the changed part is described as follows:

$$\begin{cases}\mathbf{W}=\underset{\mathbf{W}}{\operatorname{argmin}}\ \lambda||\mathbf{W}||_1+\dfrac{\eta}{2}||\mathbf{L}-\mathbf{W}-\mathbf{Q}||_{\mathrm{F}}^2 & (25)\\ \mathbf{L}=(\mathbf{D}^T\mathbf{D}+\mu\mathbf{I}+\eta\mathbf{I})^{-1}\\ \qquad\times[\mathbf{D}^T(\mathbf{X}-\mathbf{S})+\mu(\mathbf{J}-\mathbf{P})+\eta(\mathbf{W}-\mathbf{Q})] & (26)\\ \mathbf{Q}\leftarrow\mathbf{Q}-\eta(\mathbf{L}-\mathbf{W}). & (27)\end{cases}$$

1) W-Block: According to the sparse part in the ADMM as shown in (25), it is updated by applying the soft-thresholding operator as (14)

$$\mathbf{W}^{(k)}=\mathbf{softNet}_{\mathcal{R}_{\lambda/\eta}}(\mathbf{L}-\mathbf{Q}^{(k-1)}). \quad (28)$$

Similarly, the operator of the problem (28) relative to the $\mathbf{W}$ is

$$\frac{\mathbf{L}-\mathbf{Q}^{(k-1)}}{|\mathbf{L}-\mathbf{Q}^{(k-1)}|}\mathrm{ReLU}(|\mathbf{L}-\mathbf{Q}^{(\mathrm{k}-1)}|-\lambda/\eta). \quad (29)$$

2) L-Block: The objective function for $\mathbf{L}$ is written as (26), which can be expressed in the kth neural network stage using linear operation

$$\mathbf{L}^{(k)}=\mathbf{LargNet}(\Lambda,\Gamma,\Theta) \quad (30)$$

Algorithm 2 LRR-Net-S

Input: Observed HSI $\mathbf{X}$, $\mathbf{D}$, $\mathbf{L}_0$, $\mathbf{S}_0$, and parameters α, β, μ, η.
Output: Background $\mathbf{L}$, Anomaly map $\mathbf{S}$.
Initialization: $\mathbf{W}=\mathbf{0}$, $\mathbf{J}=\mathbf{0}$, $\mathbf{P}=\mathbf{0}$, $\mathbf{Q}=\mathbf{0}$, $\Lambda=\mathbf{0}$, $\Gamma=\mathbf{0}$, $\Theta=\mathbf{0}$, $\alpha=0.5$, $\lambda=1.2\times10^{-5}$, $\beta=10^{-5}$, $\mu=1$, $\eta=1$.
while $0\le k\le K$ *or meet the convergence conditions* **do**
  Fix other variables to update $\mathbf{S}$ by Eq. (14)
  $$\mathbf{S}=\mathbf{softNet}_{\mathcal{M}_\beta}(\mathbf{X}-\mathbf{DL}).$$
  Fix other variables to update $\mathbf{J}$ by Eq. (17)
  $$\mathbf{J}=\mathbf{softNet}_{\mathcal{D}_{\alpha/\mu}}(\mathbf{L}-\mathbf{P}).$$
  Fix other variables to update $\mathbf{W}$ by Eq. (28)
  $$\mathbf{W}=\mathbf{softNet}_{\mathcal{R}_{\lambda/\eta}}(\mathbf{L}-\mathbf{Q}).$$
  Fix other variables to update $\mathbf{L}$ by Eq. (30)
  $$\mathbf{L}=\mathbf{LargNet}(\Lambda,\Gamma,\Theta).$$
  Fix other variables to update penalty parameters $\mathbf{P}$ and $\mathbf{Q}$ by Eqs. (22) and (31)
  $$\mathbf{P}=\mathbf{Linear}(P_\mu),\quad \mathbf{Q}=\mathbf{Linear}(Q_\eta).$$
  Check the convergence conditions:
  if $\|\mathbf{L}-\mathbf{J}\|_{\mathrm{F}}<\varepsilon$, $\|\mathbf{L}-\mathbf{W}\|_{\mathrm{F}}<\varepsilon$, $||x^k-\hat{x}^{k-1}||_F^2<\zeta$ **then**
    Stop iteration;
  else
    $k\leftarrow k+1$;
  end
end

where $\Lambda=(\mathbf{D}^T\mathbf{D}+\mu\mathbf{I}+\eta\mathbf{I})^{-1}\mathbf{D}^T$, $\Gamma=(\mathbf{D}^T\mathbf{D}+\mu\mathbf{I}+\eta\mathbf{I})^{-1}\mu$, and $\Theta=(\mathbf{D}^T\mathbf{D}+\mu\mathbf{I}+\eta\mathbf{I})^{-1}\eta$.

3) Q-Block: As (22), the penalty parameter of $\mathbf{P}_w$ can be given by problem can be expressed as

$$\mathbf{Q}^{(k)}=\mathbf{Linear}(Q_\eta). \quad (31)$$

Similar to (22), Q_η can be calculated by $\mathbf{Q}-\eta(\mathbf{L}-\mathbf{W})$ with $\mathbf{Q}$ is also a learnable parameter offering additional flexibility, and the parameter η is updated by $\min(\max\eta,\beta*\eta)$. Such a learnable parameter plays the role of a step size that can be further adjusted to adapt to the characteristics of the detection problem. The step-update component of the LRR-Net-S network is shown in Fig. 3.

D. Initialization and Training

Typically, neural networks are trained by randomly initializing their parameters. However, to accelerate the training process, we suggest an approach of initializing the network parameters using the original parameters associated with the ADMM algorithms as follows.

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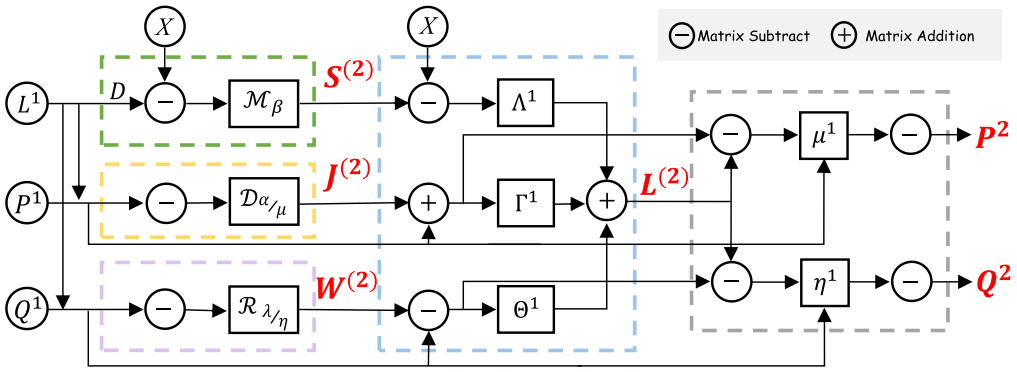

Fig. 3. Illustrating the details for the second stage network framework in LRR-Net-S. The above flowchart is equivalent to the second stage in LRR-Net with the same input–output, and operation symbols as Fig. 2.

1) The parameters Λ^k and Γ^k corresponding to the L-update component of each layer are initialized using (21).
2) The parameter $(\lambda/\eta)^k$ corresponding to the W-update component of each layer is initialized, where λ and η relate to parameters associated with the ADMM algorithm. Similar to the parameter $(\alpha/\mu)^k$.
3) The parameter β^k corresponding to the S-update component of each layer is initialized, which is closely related to the parameter of the previous step, constraining the anomalous background.
4) Finally, the parameter μ^k is associated with the J-update in the ADMM algorithm. for initialization purposes. We set the parameter to be equal to one.

During the iterative update phase of our deep unfolding net, we apply soft thresholds to iteratively update the outlier estimates. Throughout this process, all parameters of the algorithm remain learnable.

The proposed network can be trained in a completely unsupervised manner by means of standard mean-squared error (mse), similar to the well-known autoencoder. The loss function is given by

$$\mathrm{R}_{\mathrm{loss}} = \mathbf{MSE}(\mathbf{X}, \hat{\mathbf{X}}) = \frac{1}{N}\sum_{i=1}^{N}\|\mathbf{x}_i - \hat{\mathbf{x}}_i\|_2^2 \tag{32}$$

where $\mathbf{X}$ consists of N reflectance spectra $\mathbf{x}_i$, $\hat{\mathbf{x}}_i$ corresponds to the estimated value of $\hat{\mathbf{X}}$, which can also be considered as the reconstructed values.

IV. Experiments

This section demonstrates the effectiveness of the LRR-Net in two ways. First, to show the generalizability of the proposed method, we tested the proposed network using nine publicly available datasets and compared them with the nine currently popular methods. Second, the comparison of two versions of LRR-Net and LRR-Net-S demonstrates the scalability of LRR-Net. Furthermore, we analyzed the sensitivity of the two versions to the number of network layers.

A. Dataset Description

1) AVIRIS San Diego Dataset: It was captured by the Airborne Visible/Infrared Imaging Spectrometer (AVIRIS[1]) sensor from SanDiego. After removing the low signal-to-noise bands, there are 186 spectral bands used for experiments, which consist of 100 × 100 pixels covering the wavelength range from 400 to 2500 nm. Three aircraft with structural information are considered as anomalous targets of interest with 57 pixels.

2) HYDICE Urban Dataset: It was captured by the hyperspectral digital imagery collection experiment (HYDICE) airborne sensor over an urban area, CA, USA, in 1995. After removing the low signal-to-noise ratio bands, there are 80 × 100 pixels covering 175 spectral bands in a wavelength range of 400–2500 nm with a spectral resolution of 10 nm. The vehicles and roofs are considered as anomalies whose pixels account for about 0.24% of the entire image with 21 anomalous pixels.

3) AVIRIS Coast Dataset: It was also acquired by the AVIRIS sensor, which contains 10 000 spectral vectors with 207 channels covering a range of 450–1350 nm. The buildings of different scales are regarded as anomalies.

4) Pavia Beach Dataset: It was captured by the reflective optics system imaging spectrometer sensor with a spatial resolution of 1.3 m. There are 102 available spectral bands, and the spatial size is 150 × 150 pixels. There are 68 anomalous pixels in the scene.

5) Gulfport Airport Dataset: The dataset was acquired by the AVIRIS in 2010. It covers an area in Gulfport, Southern, MS, USA. The spatial resolution is 3.4 m, and the wavelength interval is 0.40–2.50 μm. 191 bands are selected after removing the low SNR bands. The spatial size is 100 × 100 pixels.

B. Implementation Details and Evaluation Metric

For the GRX method, a covariance matrix is first calculated based on the entire image, which is then input into each block to calculate the Mahalanobis distance of each pixel. According to the description in the original article of LRASR, the number of clusters K and the number of selected pixels P was set to 15 and 20, respectively, and the regularization parameters β and λ were set to 0.1 for all three experimental datasets. For LRX and CRD, we use the $W_{\mathrm{in}} = [3, 5, 7, 9, 11, 13, 15, 17, 19]$ and $W_{\mathrm{out}} = [5, 7, 9, 11, 13, 15, 17, 19, 21, 23]$, and the regularization parameter λ in CRD is set as $\lambda = 10^{-6}$. The auto-AD is built by an encoder and a decoder. The encoder consists of

[1]http://aviris.jpl.nasa.gov/

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TABLE I
AUC SCORES OF ROC CURVES (%) WITH NINE METHODS FOR NINE DATASET

	Global-RX	Local-RX	CRD	RPCA	LRASR	CNN	Auto-AD	LRR-Net	LRR-Net-S
AVIRIS-I	94.03	96.92	92.67	92.95	93.67	94.22	99.02	98.87	**99.39**
AVIRIS-II	91.10	95.67	87.65	92.74	90.59	94.67	97.20	98.13	**99.07**
HYDICE	98.57	94.92	95.31	92.55	98.53	93.06	99.10	99.36	**99.51**
Urban	98.91	99.03	93.02	97.63	91.78	98.05	99.27	99.39	**99.53**
Beach	97.93	95.69	94.31	80.67	79.31	91.36	97.85	95.32	**99.01**
Airport-1	82.21	95.76	90.73	78.39	79.32	89.49	90.29	94.35	**95.87**
Airport-2	95.26	95.37	96.09	96.27	96.89	96.97	97.32	98.67	**98.72**
Airport-3	92.88	95.10	89.04	86.78	88.91	92.16	96.75	98.13	**98.51**
Airport-4	94.03	92.63	89.96	84.31	87.58	95.02	96.17	96.89	**97.69**
Average	93.88	95.68	92.09	89.14	89.62	93.89	97.00	97.68	**98.59**

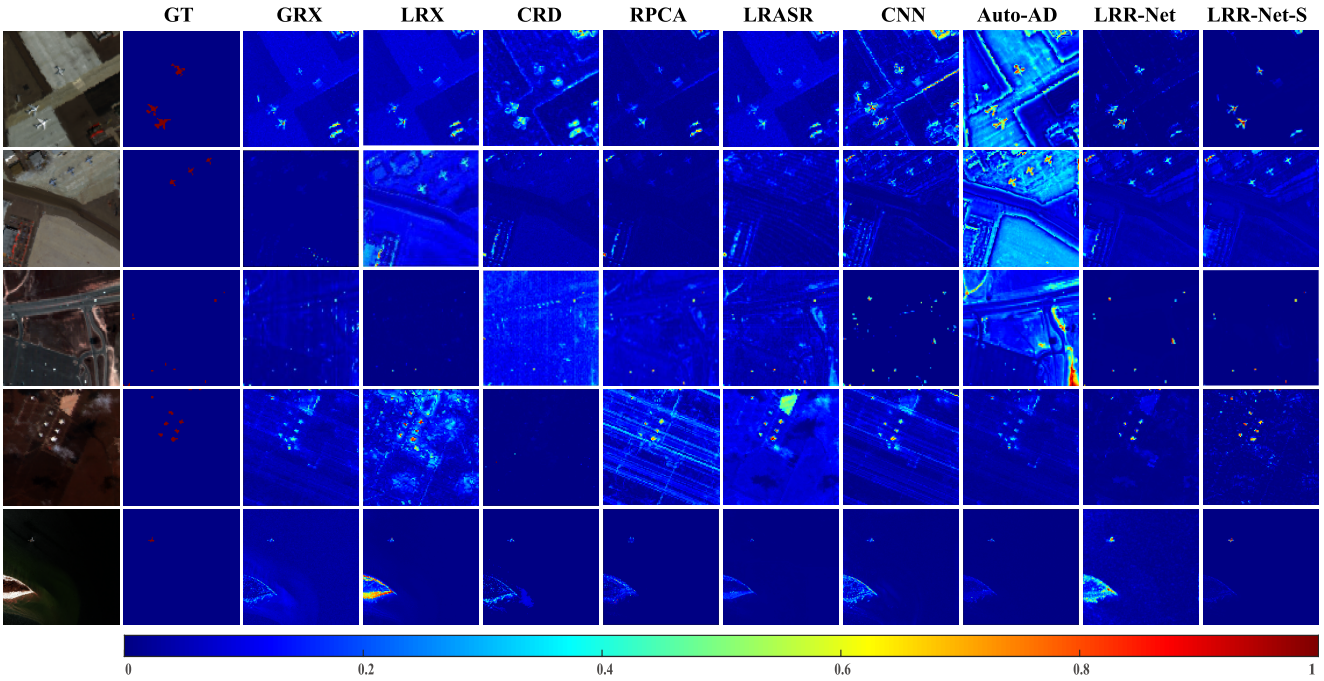

Fig. 4. Visual quality comparison on four typical scenes in HSI datasets. The pseudocolor image, ground truth, and detection map of the compared methods for AVIRIS_I, AVIRIS_II, HYDICE, Urban, and Beach are shown sequentially.

four 1-D convolution layers with the size $[C, H, W] = [64, 1, 9]$, $[128, 1, 9]$, $[256, 1, 9]$, $[512, 1, 9]$, a 1-D max-pooling layer between the convolution layers (pooling size and stride of 2), and a fully connected layer. The decoder has a structure symmetrical to the encoder, built by several deconvolutional layers.

In the simulations, the proposed LRR-Net is implemented using the Pytorch framework on Windows 10 operating system. The network is trained using Adam optimizer with a 3.20-GHz 12th Gen Intel[2] Core[3] i9-12900KF (24 CPUs) central processing unit and an NVIDIA GeForce RTX 3070 graphics processing unit (GPU). The proposed networks are trained with the following hyperparameters: the learning rate is 0.001, the batch size is set to 15, the number of updating layers is 6, and the optimization steps before the early stop when the valid loss does not descend any longer are 500.

C. Experimental Analysis and Comparison

In this section, four experiments are conducted to evaluate the performance of our proposed method. Several HS AD algorithms are compared with LRR-Net and LRR-Net-S, including classic GRX [7], LRX [12]), RPCA [46], LRSAR [16], LSMAD [47], CRD [17], Auto-AD [20], and CNN [48]. For the evaluation aspect of the computed results, color detection maps are provided in all experiments, with the colors ranging from blue to red. The receiver operating characteristic (ROC) curve [49] is given to reflect the compromise between the probability of detection (PD) and false alarm rate (FAR) shown in Fig. 6. The area under the ROC curve, namely AUC [50], is presented to quantitatively compare different algorithms. We report the AUC results in Table I and show the detection maps in Figs. 4 and 5. The colors of the detection maps range from dark blue to bright red. The regions that tend toward bright red are more likely to be anomalies.

As shown in Table I, LRR-Net (LRR-Net-S) achieves the best performance according to the AUC and ROC values. In terms of the total mean values, LRR-Net surpasses Global-RX, LRX, CRD, RPCA, LRASR, CNN, and auto-AD by 4.91%, 2.00%, 4.59%, 6.54%, 8.02%, 4.35%, and 0.68%, respectively. LRR-Net-S shows 0.91% superiority compared with LRR-Net regarding the average AUC value for all scenes. According to the detection maps of LRR-Net and LRX of

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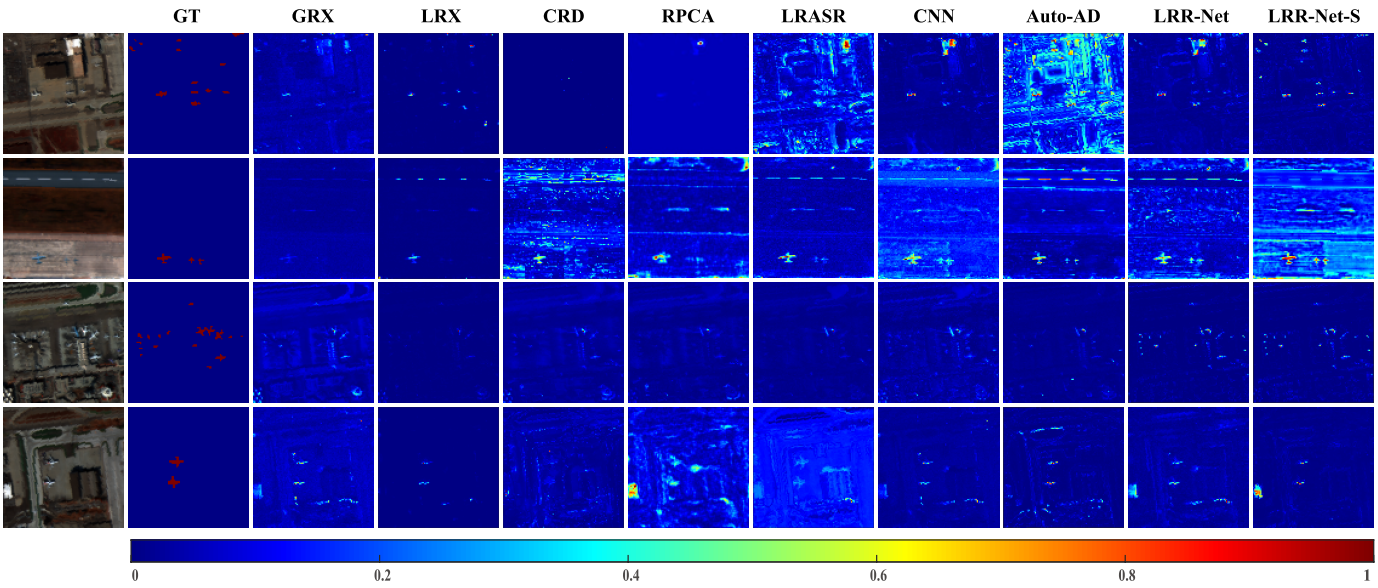

Fig. 5. Visual quality comparison on four Airport scenes for HAD. The pseudocolor image, ground truth, and detection map of the compared methods are shown, respectively.

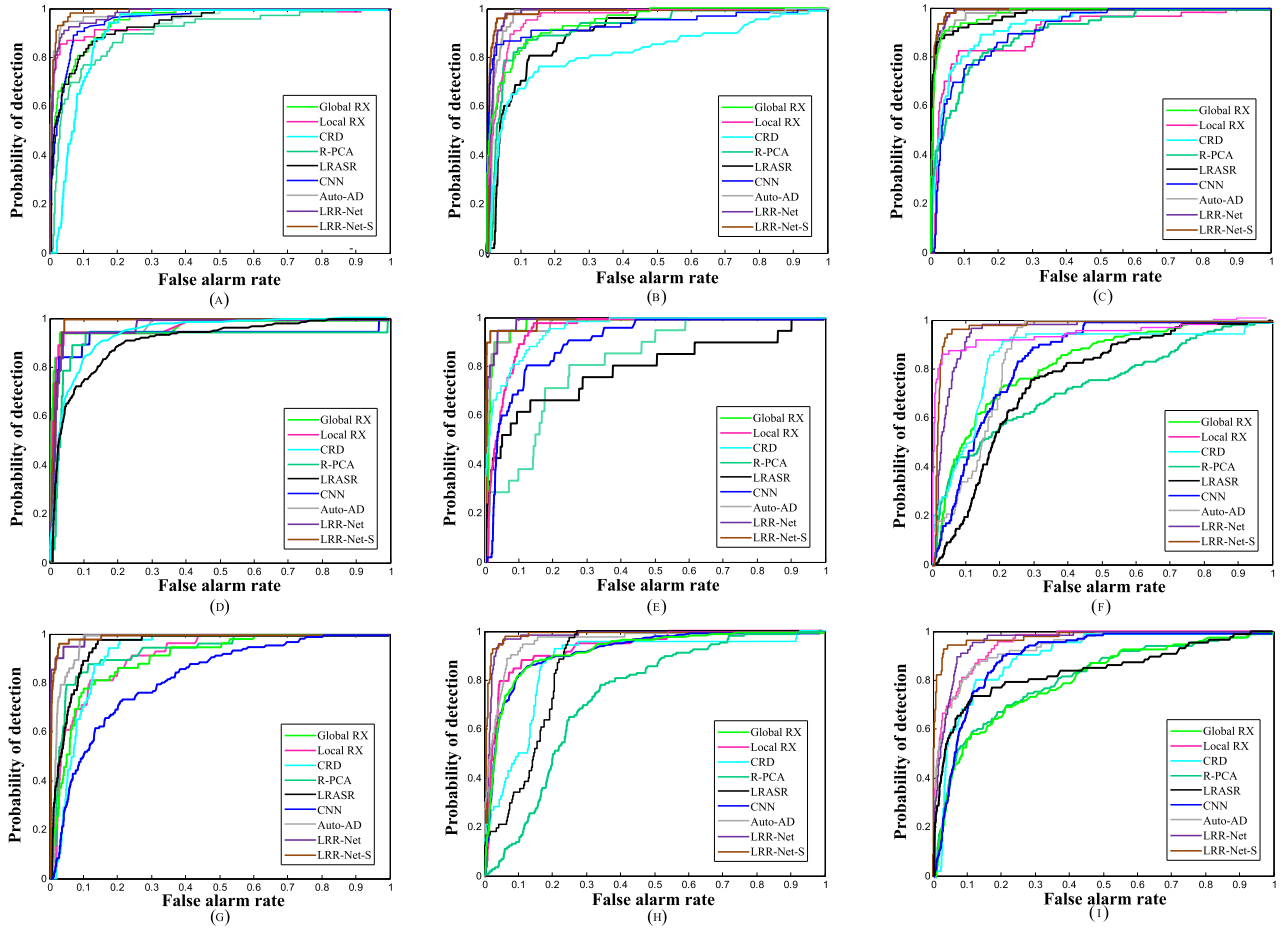

Fig. 6. Comparison for ROC curves of different methods on (A) AVIRIS_I, (B) AVIRIS_II, (C) HYDICE, (D) Urban, (E) Beach, (F) Airport-1, (G) Airport-2, (H) Airport-3, and (I) Airport-4 datasets, respectively.

the airport dataset, LRR-Net shows clearly better visualization results for the scenes Airport-2 and Airport-4. For the scenes of Beach, LRR-Net achieves significant visualization results compared with other methods.

Compared to traditional model-based approaches, the GRX algorithm exhibits poor performance in detecting anomalies in complex background scenes. This is because it neglects local information by relying solely on the square value of

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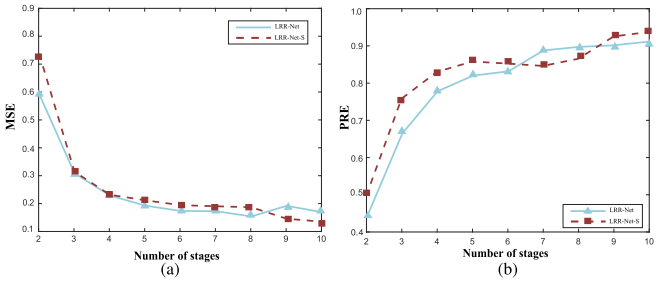

Fig. 7. Achievable detection performance of LRR-Net with respect to a number of updating stages: (a) MSE and (b) PRE.

the Mahalanobis distance between the image to be detected and the mean of the global background. On the other hand, the CRD method uses a sliding dual window to approximate each pixel's representation as a linear combination of surrounding samples, thus better utilizing both global and local background information and achieving improved separability between background and target. However, these model-based approaches still struggle to resist interference information such as shadows, trees, and edges in complex scenes due to their limited representation ability. Moreover, traditional detectors may fail to reflect the effects of certain data accurately, and their detection results can be highly volatile.

Due to the powerful learning ability of deep networks, AE-like models have relatively strong robustness in datasets with varying complexity. However, due to the lack of necessary prior knowledge, these models tend to get trapped in local solutions despite their strong data-fitting ability. This may explain why AE fails to effectively detect anomalies in hyperspectral images with complex background. On the other hand, our LRR-Net outperforms other algorithms, demonstrating that the unfolding network can effectively address HAD problems in an adaptive manner. Using model-based methods to guide the search for optimal parameters in neural networks is a more efficient and physically meaningful approach.

In summary, as the detection plots of the nine algorithms show, our LRR-Net always shows the best results for different size datasets, different backgrounds, and anomalies. The detection performance of all methods decreases as the background complexity increases, while LRR-Net has strong robustness.

D. Performance Analysis

In this experiment, we analyzed the performance indicator to evaluate the impact of the number of update stages in LRRNet on detection efficiency. The results of the experiment are presented in Fig. 7. We used two indicators, MSE and PRE, to measure performance, which is defined as follows:

$$\mathbf{MSE} = \frac{\|\mathbf{X} - \mathbf{X}_{\text{recon}}\|^2}{\|\mathbf{X}\|^2} \tag{33}$$

where $\mathbf{X}$ denotes the original image and $\mathbf{X}_{\text{recon}}$ represents the recovered image, and

$$\mathbf{PRE} = \frac{\|\mathbf{X}_S - \mathbf{X}_{gt}\|^2}{\|\mathbf{X}_{gt}\|^2} \tag{34}$$

where $\mathbf{X}_S$ denotes the anomaly image and $\mathbf{X}_{gt}$ represents the ground-truth image.

Our findings suggest that the detection performance improves notably as the number of updating stages increases, especially when the number of stages is less than nine for both experiments. This improvement can be attributed to the introduction of additional learnable parameters and updates that come with the inclusion of more updating stages. However, we also observed that detection performance only marginally improves beyond the nine stages. This implies that nine updating stages are adequate to ensure satisfactory detection performance. Therefore, the hundreds of iterations involved in the iterative joint low-rank and sparse recovery method can be substituted with the nine-stage LRR-Net, which can learn optimal parameters from the data.

V. Conclusion

In this article, we introduce a novel interpretable deep unfolding network, called LRR-Net, for the task of HAD, which establishes a connection between low-rank model-based methods and data-driven DL techniques. LRR-Net utilizes the ADMM optimizer to efficiently solve the LRR model and employs the obtained solution as prior knowledge to guide the optimization of network parameters. Moreover, LRR-Net automatically learns threshold and weight parameters in an end-to-end unsupervised manner, eliminating the need for manual parameter tuning. Additionally, we propose LRR-Net-S, which jointly recovers sparse and low-rank matrices, demonstrating the scalability of the LRR-Net framework. We conduct experiments to verify the effectiveness of our proposed method, and the results demonstrate that LRR-Net outperforms iterative HAD methods in detection performance. In our future work, we would like to further optimize the model with a focus on the dictionary design to better learn the underlying spatial structure features of the data.

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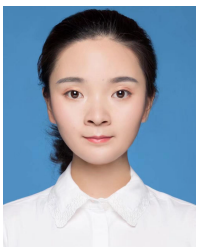

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